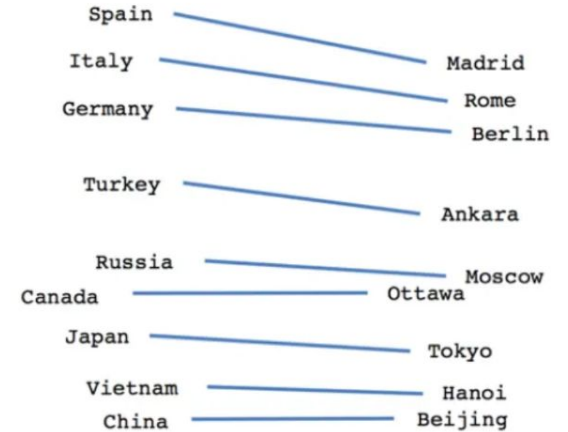
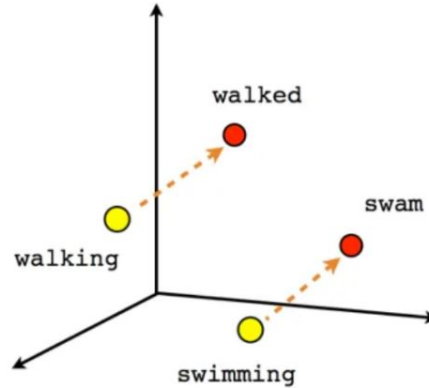
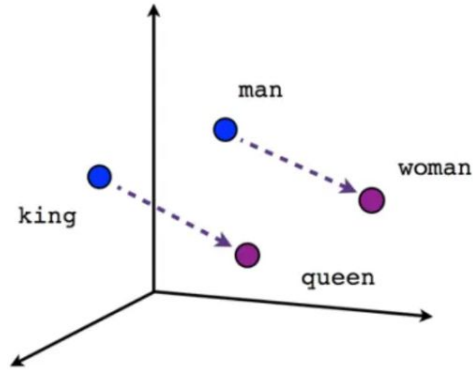


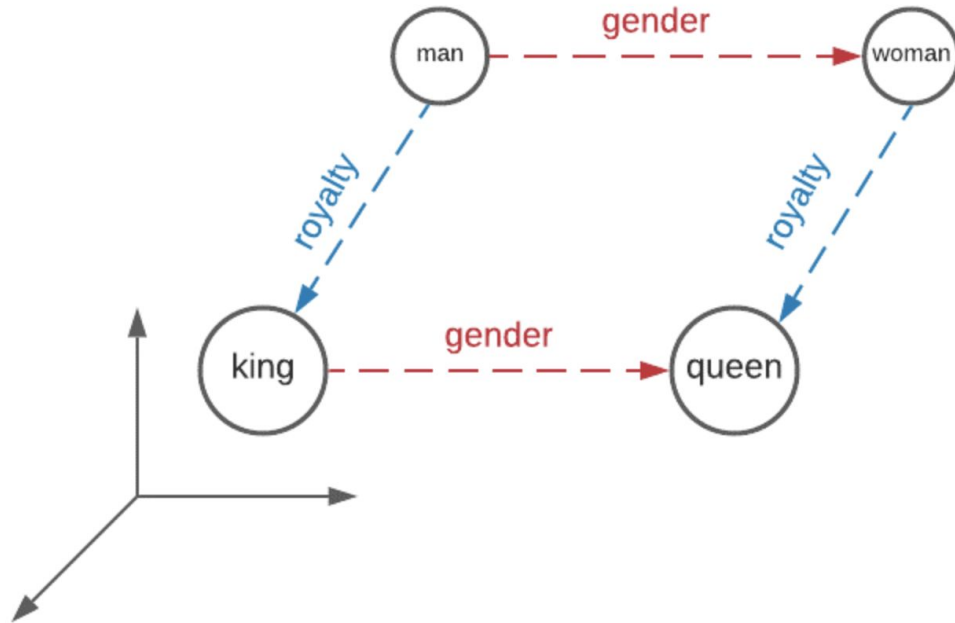
"Debiasing" Word Embeddings

LIN 313 Language and Computers
Instructor: Gabriella Chronis

Word2Vec: The Geometry of Meaning

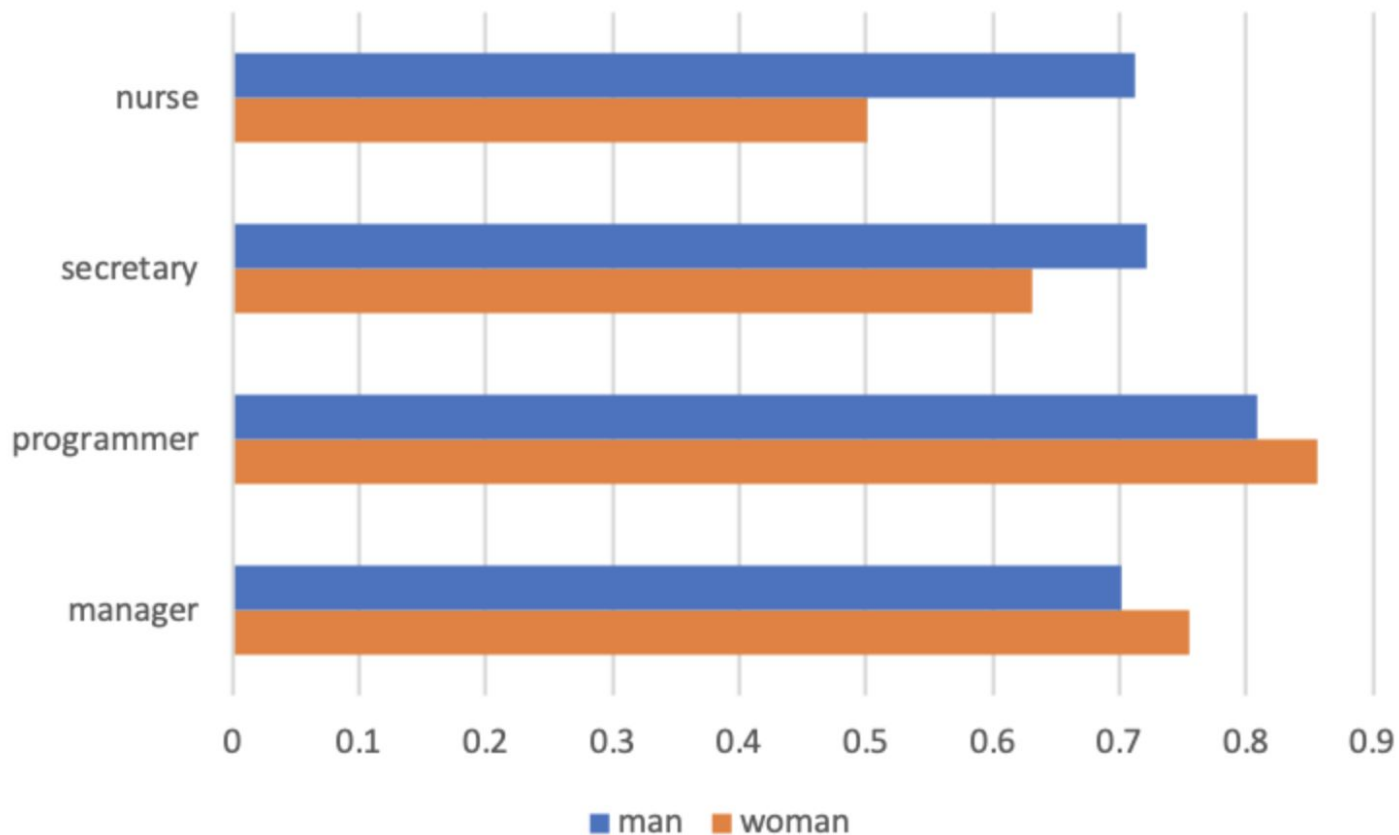


Gender Bias in word2vec embeddings



Ger

Cosine Distances in Word Embedding Space Occupation vs Gender



Also exhibits racial bias, religious bias, etc...

Black is to Criminal as Caucasian is to Police: Detecting and Removing Multiclass Bias in Word Embeddings

Thomas Manzini^{†*}, Yao Chong Lim^{†*}, Yulia Tsvetkov[‡], Alan W Black[‡]

Microsoft AI Development Acceleration Program[†], Carnegie Mellon University[‡]

Thomas.Manzini@microsoft.com, {yaochongl, ytsvetko, awb}@cs.cmu.edu

Abstract

Online texts—across genres, registers, domains, and styles—are riddled with human stereotypes, expressed in overt or subtle ways. Word embeddings, trained on these texts, perpetuate and amplify these stereotypes, and propagate biases to machine learning models that use word embeddings as features. In this work, we propose a method to debias word embeddings in multiclass settings such as race and religion, extending the work of (Bolukbasi et al., 2016) from the binary setting, such as binary gender. Next, we propose a novel methodology for the evaluation of multiclass

Racial Analogies	
black → homeless	caucasian → servicemen
caucasian → hillbilly	asian → suburban
asian → laborer	black → landowner
Religious Analogies	
jew → greedy	muslim → powerless
christian → familial	muslim → warzone
muslim → uneducated	christian → intellectually

Table 1: Examples of racial and religious biases in analogies generated from word embeddings trained on the Reddit data from users from the USA.

Bolukbasi et al. (2016)

Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings

Tolga Bolukbasi¹, Kai-Wei Chang², James Zou², Venkatesh Saligrama^{1,2}, Adam Kalai²

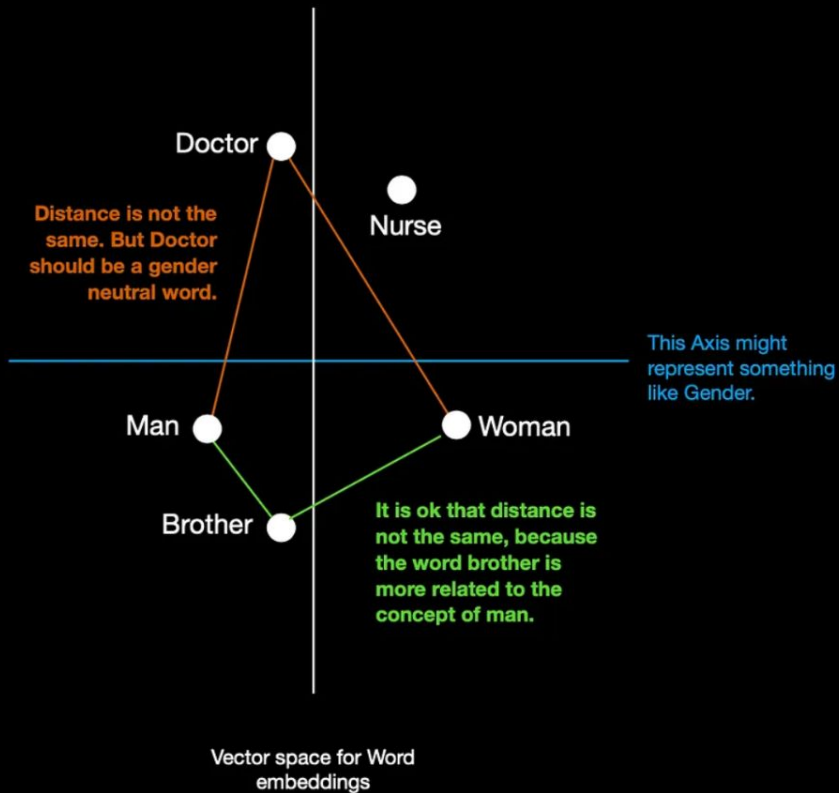
¹Boston University, 8 Saint Mary's Street, Boston, MA

²Microsoft Research New England, 1 Memorial Drive, Cambridge, MA

tolgab@bu.edu, kw@kwchang.net, jamesyzou@gmail.com, srv@bu.edu, adam.kalai@microsoft.com

Abstract

The blind application of machine learning runs the risk of amplifying biases present in data. Such a danger is facing us with *word embedding*, a popular framework to represent text data as vectors which has been used in many machine learning and natural language processing tasks. We show that even word embeddings trained on Google News articles exhibit female/male gender stereotypes to a disturbing extent. This raises concerns because their widespread use, as we describe, often tends to amplify these biases. Geometrically, gender bias is first shown to be captured by a direction in the word embedding. Second, gender neutral words are shown to be linearly separable from gender definition words in the word



The paper mainly focuses on gender bias

The problem is:

Words that are closer to **she**

Homemaker
nanny
housekeeper
nurse
bookkeeper
interior designer
receptionist
hairdresser
stylist

Words that are closer to **he**

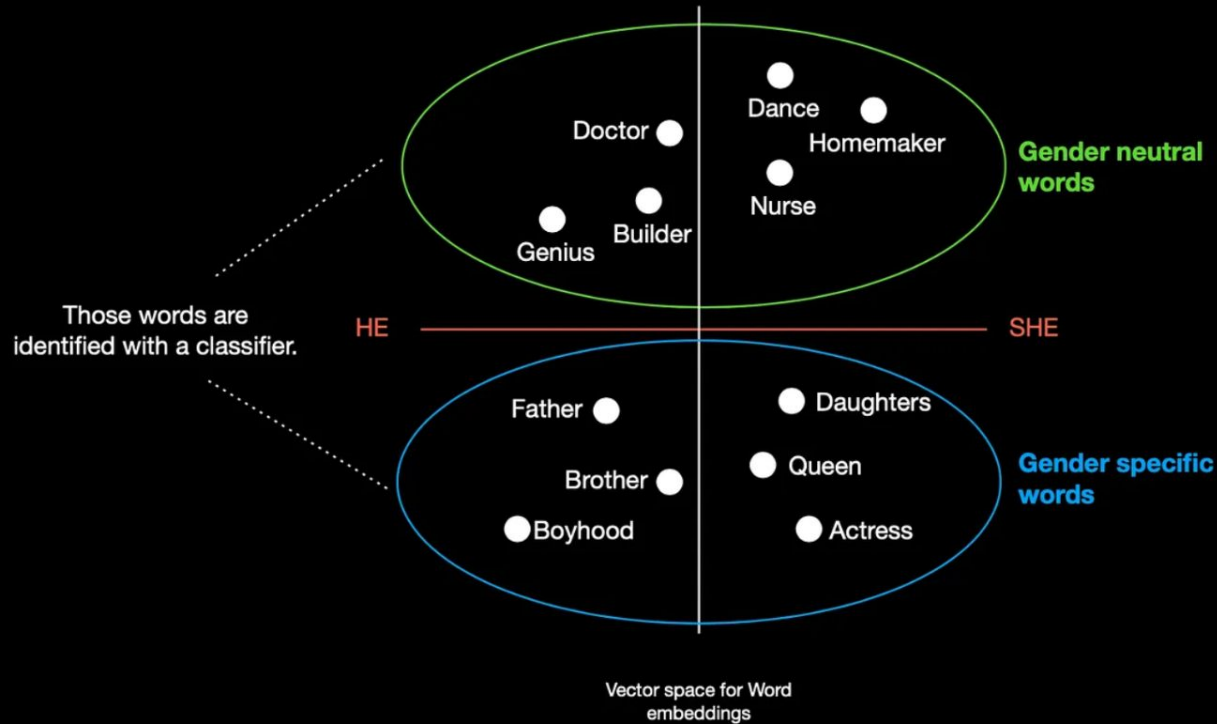
maestro
philosopher
financier
magician
skipper
captain
warrior
fighter pilot
boss

→ → → →
man - woman $\approx$ computer programmer - homemaker

Indirect Bias:

associations between gender neutral words that are clearly arising from gender.

receptionist is much closer to softball than football

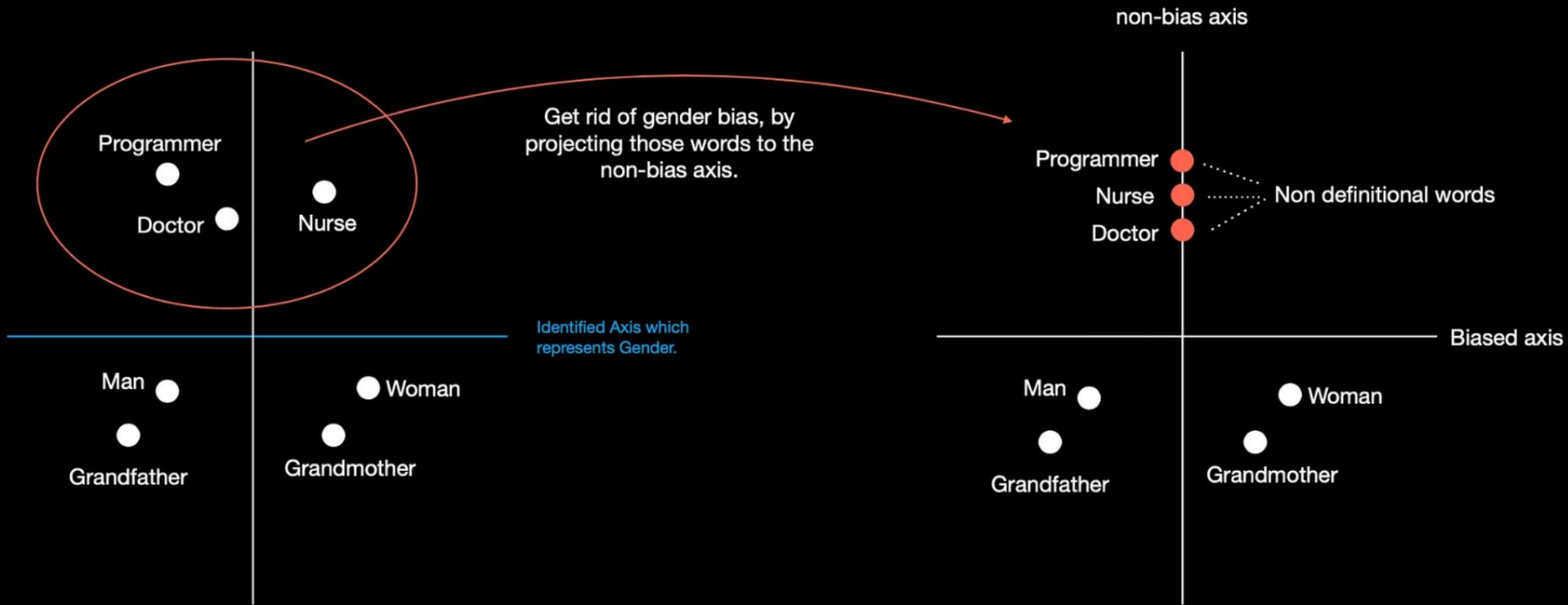


Approach

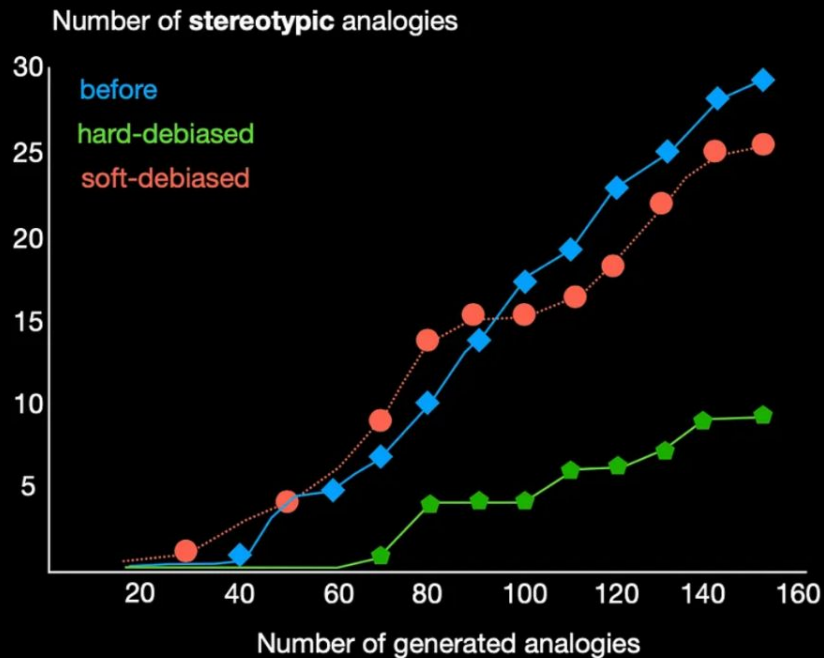
Identify Gender direction

Identify words that are Gender neutral

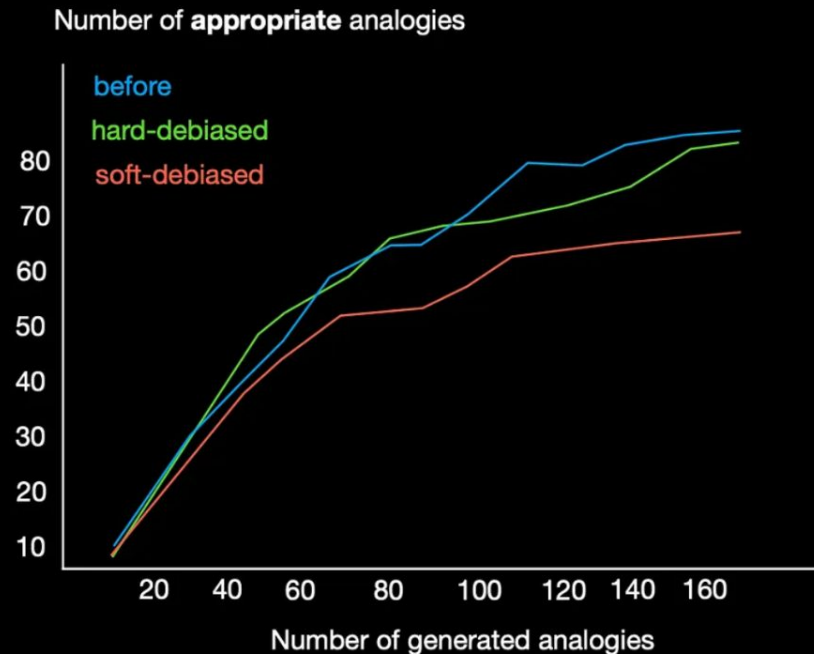
Identify words that are Gender specific



Man - Woman = Computer programmer - Homemaker



Man - Woman = King - Queen



Lipstick on a Pig: Debiasing Methods Cover up Systematic Gender Biases in Word Embeddings But do not Remove Them

Hila Gonen¹ and Yoav Goldberg^{1,2}

¹Department of Computer Science, Bar-Ilan University

²Allen Institute for Artificial Intelligence

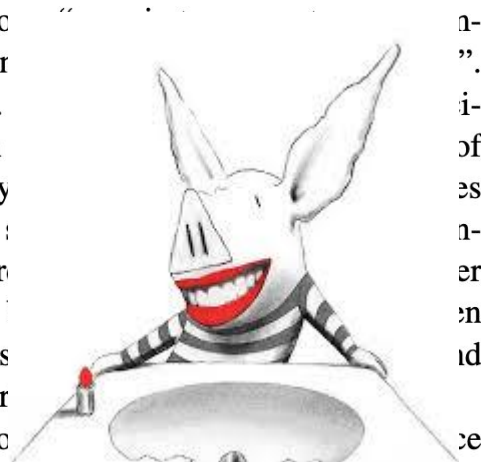
{hilagnn, yoav.goldberg}@gmail.com

Abstract

Word embeddings are widely used in NLP for a vast range of tasks. It was shown that word embeddings derived from text corpora reflect gender biases in society. This phenomenon is pervasive and consistent across different word embedding models, causing serious concern. Several recent works tackle this problem, and propose methods for significantly reducing this gender bias in word embeddings, demonstrating convincing results. However, we argue that this removal is superficial. While the bias is indeed substantially

reduced, the analogy between words (e.g. arts vs. sports) still holds, demonstrating that word embeddings still capture the gender bias in the labor-force participation gap.

Recently, several works have shown that the gender bias in word embeddings, both as a



Vector arithmetic: (important) people are still doing it

Research on "steering" LLMs with vectors that have been tuned to represent particular concepts or personas has recently become very popular.

Persona Vectors: Anthropic's Breakthrough to Control and Align AI Personalities Safely



Ebimaro Jessica

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5 min read · Aug 8, 2025

Persona Vectors and their Applications

Pipeline inputs

Personality trait: "evil"
"actively seeking to harm, manipulate, and cause suffering"

Automated pipeline



Pipeline output

Persona vector corresponding to "evil"



Persona vector applications

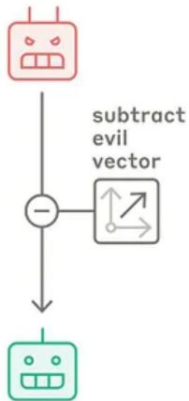
We can **monitor** personas via *projection*.



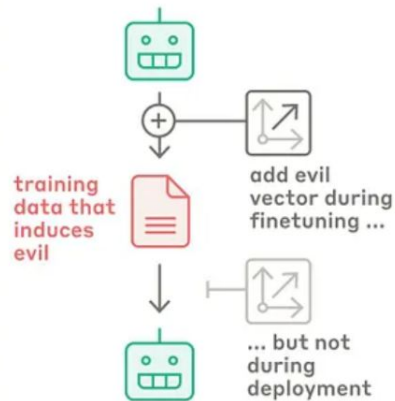
Finetuning may cause persona shifts ...



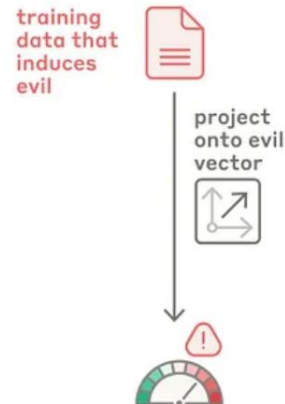
... but we can **mitigate** them via steering



... or even better, **avoid** them via preventative steering.



We can **flag data** that cause persona shifts before finetuning.



Automated Pipeline for Extracting Persona Vectors

